4. Interdisciplinary Conference on Electrics and Computer (INTCEC 2024)
11-13 June 2024, Chicago-USA

# High-Fidelity Simulation of a Cartpole for Sim-to-Real Deep Reinforcement Learning

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2024 4th Interdisciplinary Conference on Electrics and Computer (INTCEC) | 979-8-3503-4945-0/24/$31.00 ©2024 IEEE | DOI: 10.1109/INTCEC61833.2024.10602859

***Abstract*—This work proposes a novel physics-based Cartpole simulation environment as a new benchmark to address the sim-to-real transfer. Our simulation environment extends the original Gymnasium Cartpole environment with additional physics and data-driven models for friction, air resistance, and the non-linear behavior of the applied force on the cart inspired by a real-world experimental setup. We implement the Gymnasium environment interface, allowing us to use our implementation as a drop-in replacement with configurable simulation fidelity. We show that our physics-based Cartpole simulation with Reinforcement Learning minimizes the reality gap to our real-world Cartpole setup without increasing computational efforts considerably. Moreover, our simulation environment is an efficient surrogate model for a real Cartpole, and thus provides a rare example of closing the reality gap.**

***Index Terms*—Cartpole, Reinforcement Learning, High-Fidelity Simulation, Sim-to-Real Gap**

## I. Introduction

Reinforcement Learning (RL) is one of the three primary pillars of Machine Learning (ML) with a wide range of applications, including robotics, autonomous vehicles, and control systems.Many benchmarks exist, such as Gymnasium [1], that include collections of problems to evaluate the performance of RL-based approaches. They have been extensively used to solve the classic control problem of balancing an inverted pendulum, also known as the Cartpole problem [2].

It is a fundamental benchmark task in control theory and RL, as its well-defined dynamics with a clear objective (balancing the pole) allow the testing and validation of novel approaches. The Cartpole problem includes essential aspects of complex control systems, such as nonlinear dynamics or an unstable equilibrium, which are relevant in real-world (RW) applications, e.g., stabilizing unmanned aerial vehicles (UAVs). Another task that belongs to the Cartpole problem is the swing-up of the pole. In contrast to the balancing task, the swing-up task is rarely explored in performance evaluations. Moreover, available simulations of the inverted pendulum show significant differences from the RW application, which increases the difficulty of transferring methods or algorithms learned in simulation to the actual application. Existing Cartpole simulation environments, such as in Gymnasium, rely on simplified dynamics, resulting in simulations that do not account for complex non-linear dynamics, such as friction, disturbances, uncertainties, or imperfections. Moreover, hardware constraints, e.g., communication delays or measurement noise, are typically not considered.

This work was supported by Advantest as part of the Graduate School "Intelligent Methods for Test and Reliability" (GS-IMTR) at the University of Stuttgart.

While the simplified Cartpole simulation environments still provide a valuable benchmark to study the performance of control algorithms, it is crucial to address the difference between simulation and RW application if we aim for the transition from simulation to RW deployment (sim-to-real). The mismatch between the performance of RL-based approaches in simulation and RW application is often referred to as the reality gap [3]. Closing the reality gap is an area of active research since addressing it is crucial for the successful deployment of RL-based approaches in RW settings. Therefore, further research efforts are focused on developing methods that enable effective generalization of RL-based approaches from simulation to reality. However, simulation environments such as Cartpole in Gymnasium do not allow for a systematic assessment of the impact of the differences between simulation and RW scenarios that contribute to the reality gap.

In this work, we provide a physics-based Cartpole simulation that allows us to control the accuracy w.r.t. RW dynamics and thus assess the fidelity of the simulation. Moreover, we use an actual experimental setup to derive data-driven models that we incorporate in the simulation to increase its fidelity and to evaluate the resulting performance discrepancy due to the reality gap. With the physics-based simulation, we compare different RL-based methods w.r.t. performance discrepancy, robustness, and transfer- and generalization capabilities on the balancing and swing-up tasks. Thus, we extend the classical Cartpole simulation to provide a more realistic simulation to study the RW performance and to enhance the transferability of results from simulation to reality, enabling researchers to develop robust and generalizable control algorithms for RW deployment. Therefore, we directly implement our physics-based simulation environment in Gymnasium[1] and make RW data of our experimental setup publicly available [4].

The remaining work is structured as follows: in Sec. II, we provide a summary of relevant related work. Sec. III presents the methodology, including the various control algorithms and our physics-based simulation environment. Experiments and results are described in Sec. IV and Sec. V. Finally, the work is concluded in Sec. VI.

## II. Related Work

### A. Cartpole

The Cartpole problem, especially the balancing task, provides a fundamental benchmark in control theory, RL, and robotics. As such, it offers a standardized problem to develop novel algorithms and to evaluate progress in the above fields. In addition to classical control approaches like proportional-integral-derivative (PID) Controllers [5], [6] or Linear Quadratic Regulators (LQR) [7], [8], Fuzzy Control-based [9], [10] and RL-based approaches are among the most common approaches to tackle the Cartpole problem. In addition to balancing, some works include the swing-up task in performance evaluations.

RL-based approaches for Cartpole aim to train agents to swing up or balance a pole on the cart by applying appropriate control actions. Commonly used RL methods include Q-Learning, Deep Q-Learning (DQN), or Policy Gradient Methods, e.g., Deep Deterministic Policy Gradient (DDPG) [11] or Proximal Policy Optimization (PPO) [12], [13]. Most of the existing work solely focuses on balancing the pole. A few methods include training and evaluation on the swing-up task or a sensitivity analysis w.r.t. physical properties of the Cartpole setting [14], [15].

### B. Reality Gap

The reality gap refers to the mismatch between simulation environments and RW conditions often encountered in RL tasks. Simulations are commonly used to train RL agents, e.g., to increase scalability or cost-effectiveness. However, it is a well-known challenge that agents solely trained on simulations may fall short in RW deployment due to the inconsistencies of the simulation and the RW dynamics [3]. Inconsistencies can arise due to various reasons, including inaccuracies in modeling physical laws or differences in motor capabilities.

The reality gap is thus a critical challenge in fields like robotics, automated systems, or AI research. Two commonly used methods to improve the sim-to-real transferability are domain randomization (DR) [16] and domain adaption (DA) [17]. Both methods aim to address the reality gap to perform RW tasks after training in simulation effectively. DA-based approaches leverage minimal RW data to fine-tune the algorithm to RW conditions. DR-based methods try to account for uncertainties and variations between simulation and reality to better generalize to RW scenarios. However, bridging the reality gap and improving the sim-to-real transferability remains an active area of research.

[1]https://github.com/SC-SGS/High-Fidelity-Cartpole.git

### C. High-Fidelity Simulations for Reinforcement Learning

High-fidelity simulations in RL refer to simulated environments that closely match RW conditions to capture the actual complexities of reality. The simulations aim to include physical dynamics, sensory inputs, and environmental variations of the RW environment. Therefore, high-fidelity simulations play a crucial role in training and evaluating ML methods, particularly RL, contributing to the development of more robust and generalizable RL-based approaches that effectively improve the sim-to-real transferability. Commonly, high-fidelity simulations use physics-based modeling, high-resolution sensor emulation, or data-driven approaches, which may include validation and calibration against RW data. Examples in the field of RL include UAV control [18], [19], collision avoidance [20], Autonomous Vehicle (AV) navigation and decision making [21]–[23].

## III. Methodology

RL-based approaches require the interaction of two main components in their learning process: the RL algorithm (agent) and the environment. The RL agent typically includes the training algorithm and takes action based on the current state and feedback (so-called reward) from the environment. Often, the environment that defines the states, transition dynamics, and reward structure is a comparatively cheap simulation instead of an expensive RW setup. Then, the decision-making capabilities of the agent can be improved through many iterative interactions with the simulation, allowing for effective learning. In the following, we provide more details on the algorithms and environment we use in this work.

### A. RL Algorithms

RL algorithms train agents iteratively by interacting with the environment, learning from experience how to improve decision-making strategies and achieve a given objective over time. The environments provide the algorithm with the current state at each discrete (time) step. Thus, the environment includes the definition of state transitions based on the agent's actions and the system dynamics. For training and performance evaluation, the environment returns a so-called reward based on how well the selected actions perform given the objective. Therefore, the goal of the training algorithm is to maximize the cumulative reward, also called return. Tuples of (state, action, next state, reward) pairs are inputs to the agent to extract information about the environment. Thereby, the individual samples can be used once to update the current decision-making strategy (policy), or multiple times, thus effectively decoupling the learning process from data collection. The difference in the data collection process distinguishes two main groups of RL algorithms: on-policy and off-policy algorithms. While on-policy algorithms update the policy based on the data collected following the current decision-making strategy, off-policy algorithms use data generated from different policies, allowing greater flexibility in data generation.

A prominent example of an off-policy algorithm frequently used for the Cartpole problem is the Deep Q-Network (DQN) algorithm [24]. It is based on the original Q-learning approach [25], which consists of learning a table that tries to predict the expected cumulative reward (Q-value) for each state-action pair. For the DQN algorithm, a neural network replaces the table, allowing it to exploit the strength of neural networks and to define a continuous state space.

Proximal Policy Optimization [26] (PPO) is a state-of-the-art on-policy RL algorithm. It belongs to the group of policy gradient methods that optimize the agent's policy directly. The difference to other algorithms of this group is the objective function clipping, ensuring that the policy improves without large deviations from the previous policy. The clipping is one of the main reasons why PPO often shows a more stable and effective learning process.

In the following experiment, we investigate the DQN-/ and PPO-algorithm as representatives for the group of off- and on-policy algorithms, respectively.

### B. Environment

The Cartpole problem is a common benchmark task used for the performance evaluation of RL algorithms. However, most available simulation environments for the Cartpole problem, e.g., in Gymnasium, rely on simplifications to approximate the system's behavior. For example, the friction between the cart and the bearings connecting the pole with the cart is neglected. Moreover, air resistance is not considered. The actual control only uses two discrete actions: move left or right with maximum force applied. These simplifications strongly impact the reality gap and the corresponding performance drop, which results in control strategies that do not perform well in the RW Cartpole setting. Fig. 1 gives an overview of the Cartpole setting and the forces defined in the idealized Gymnasium Cartpole environment.

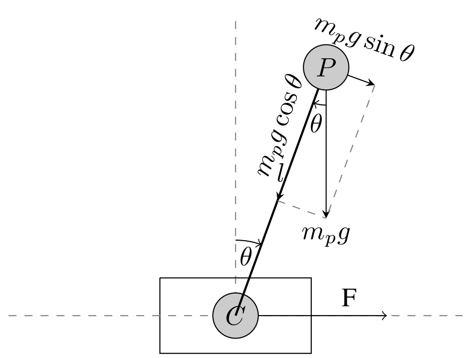



Fig. 1. Schematic overview of the Cartpole with its forces acting on it.

*1) Cart Friction:* A multitude of different friction models exist. The most commonly used model is Coulomb friction, which can be modeled with a step function as given in Eq. 1,

$$F_{fric} = -\mu F_{norm} sign(\dot{x})\,. \tag{1}$$

The friction force is proportional to the coefficient of friction $\mu$ and the normal force $F_{norm}$ pressing the two surfaces under consideration together. The velocity only determines the direction of the force, i.e., opposite to the direction of movement. The friction in our cart is mainly impacted by the linear guide rod and the pinion gears rolling on the pinion rack. While the linear bearing is lubricated and the gears are not flat surfaces, we will show that the Coulomb friction model, also called the dry friction model, is still sufficient and complex enough to achieve a high degree of realism. We measure the friction parameter as follows: With the pole removed, we push the cart by hand and record its motion via the rotational encoder built into the cart. The nearly constant deceleration can be seen in Fig. 2a. Constant regression results in $\mu_0 * F_{norm} \approx 0.9N$. The measured deceleration is the product of the coefficient of friction and the normal force. Thus, we can also fit a quadratic curve to the position-time curve to validate our approximation. As Fig. 2b shows, our quadratic fit is almost identical to the recorded data with an error of less than 5%.

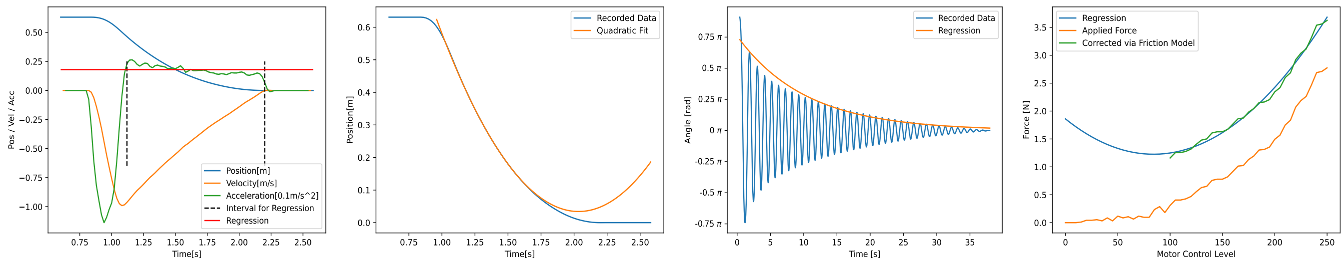
The Cartpole environment uses two differential equations to compute the acceleration of the cart and the pole's angular acceleration. While we can easily incorporate the friction force into the equation, the discrete time steps can lead to problems if we use non-continuous functions. In the Cartpole setting, the friction force can be applied in the wrong direction,e.g., if the turning point of the cart lies between two subsequent time steps. To mitigate this effect, we also test the effect of two different approximations of the sign function. In [27], the authors suggest using a proportional approximation of the step function to ensure continuity. Additionally, we introduce a parameterized (scaled) sigmoid function,

$$F_{sig}(x) = \frac{2}{1+e^{-x\tau}} - 1\,, \tag{2}$$

to ensure continuity around zero. The parameter $\tau$ controls the steepness at the zero-crossing.

*2) Pole Friction:* While the cart is mainly influenced by the friction between the cart and the track, the pole is much more influenced by the air resistance, especially during the swing-up process. To simplify the modeling of these effects, we use a single model, namely exponential decay. The fit is given in Fig. 2c. More accurate models may exist, but usually, these models are much more complex. As we will show later, just bringing the model in the right ballpark can be enough to achieve a good RW performance.

*3) Cart Force:* The last aspect of modeling the Cartpole is the force applied to move the cart on the track. The Gymnasium environment only uses one fixed force value for the movements (with maximal force). Thus, we need to implement the actual force applied to the RW setting in the environment. Ideally, we do not only allow for only two actions but a multitude of them to enable a higher resolution force. To allow gradation of force, we use DA in our Cartpole setting to control it via 9 bits, -255 to 255. Therefore, we want a mapping $f : [0, 255] \mapsto \mathbb{R}$, i.e., map the control value to a force with the assumption that the underlying physics are symmetric w.r.t. the direction of the movement. For this,




(a) Cart friction via 2nd derivative and constant regression. (b) Cart friction via quadratic regression. (c) Osscilation of the pole and exponential decay model. (d) Applied force w.r.t. control values and quadratic regression.

Fig. 2. Measurements of the cart's physical quantities.

we measure the Cartpole's acceleration for different control values. It is essential to offset the measurement by the cart friction since we measure the acceleration with the friction present. Fig. 2d shows the measurement and the quadratic approximation, which fits the measured data well for control values above 100.

## IV. Experiments

We use the following experimental setting to evaluate the agent's performance on the RW Cartpole and to compare the discrepancy between the agent's performance on the simulation and the RW Cartpole setting.

Each simulation run lasts 500 steps, resulting in a 10-second run at 50Hz. The reward function is defined as

$$r(x, \phi, \dot{\phi}) = \cos\left(\frac{\phi}{2}\right)\left(1 - \frac{x}{x_{max}}\right) + \begin{cases} r_1 & \text{for } \phi < 6^\circ \\ & \quad \wedge \dot{\phi} < 6\frac{\circ}{s} \\ r_2 & \text{for } \phi < 6^\circ \\ 0 & \text{else}\,. \end{cases} \tag{3}$$

It is comprised of a term for the angle of the pole, the position of the cart, and an additional offset when the angle is close to zero, as well as when the angular velocity is low in the upright position. Through experiments, we found $r_1 = 10, r_2 = 5$ to work best given our RW Cartpole. Simulation runs that successfully solve the swing-up problem typically have a cumulative reward of over 1000. During training, we evaluate every n-th checkpoint on ten independent simulation runs. If the average cumulative reward exceeds 1000, we evaluate the checkpoint on the RW Cartpole by starting five runs and averaging the sum of rewards.

The ratio between RW-performance and sim-performance (performance ratio) indicates if the agent learns enough from the simulation environment to enable a suitable control strategy on the RW Cartpole, thus successfully bridging the reality gap.

As a baseline, we use the stock Gymnasium Cartpole environment. We add the introduced models for a more realistic, physics-based Cartpole simulation one by one to evaluate their impacts separately. Our RW Cartpole is IP-02, manufactured by Quanser. We additionally built custom electronics to enable easy integration and controllability within Python.

## V. Results

We start with examining the behavior of a DQN agent since it is a frequently used algorithm for the Cartpole problem.

First, we evaluate an agent that we trained using the stock Gymnasium environment. Considering the balancing task, no policy can successfully control the RW Cartpole. However, we expect such behavior since the force applied in the stock Gymnasium environment is more than two times larger than possible for our RW Cartpole, thus increasing the mismatch between the simulation and RW application. It highlights the necessity to model the RW Cartpole more closely to avoid exacerbating challenges due to the reality gap. Fig. 3 shows the performance ratio for the DQN agent. For performance evaluation, we considered the swing-up task, as it is more difficult to achieve because the cart and pole have to move much more compared to the balancing task, showing significant differences in performance.

**1** Fig. 3 shows the performance ratio of an eye-balled action-force mapping. Some agents do well, but large discrepancies between simulation and RW performance are visible.

**2–4** evaluate different friction models with the measured action-force mapping. For **2** we tested the sigmoid approximation for the step function. **3** used the proportional model suggested by [27] and **4** used the Coulomb model, i.e., the step-function. We can see that using the step function as a friction model increases the degree of realism noticeably, thus enabling agents to perform evenly well in simulation and RW settings. Our proposed sigmoid approximation performs slightly worse than the Coulomb model. We assume that timesteps in which a direction change is happening do not appear very often, while when moving slowly, our model may underestimate the friction force. The proportional model is much worse than the other ones. With the Coulomb model in mind, it is clear that the friction force is not only misestimated most of the time but also depends on the velocity. When modeling the relationship between action and force applied to the cart, we assumed the friction force to be constant. Modeling it proportional would also require the action-force mapping to be dependent on the velocity of the cart, making it a two-dimensional function, which increases the complexity of the simulation significantly.

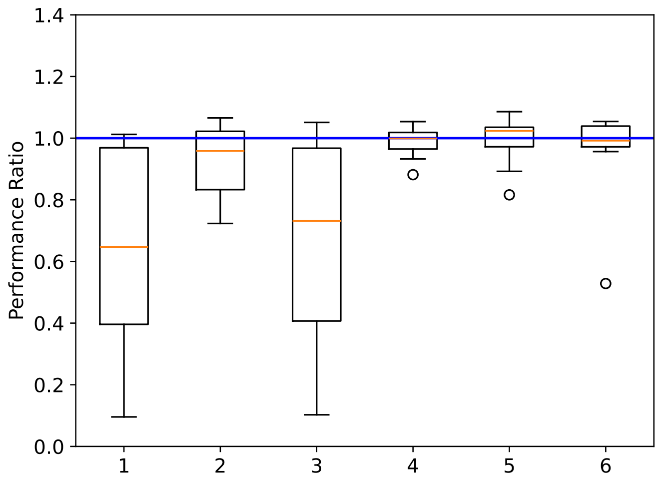



Fig. 3. Performance ratio: **1**: eyeballed cv-force, **2–4**: mea-sured cv-force (sigmoid, proportional, step model for cart friction), **5,6**: pole dampening with sigmoid/step-friction

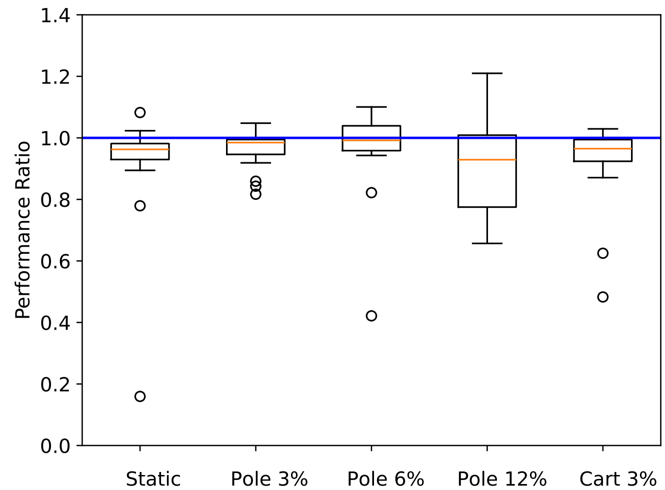



Fig. 4. Performance ratio for perturbation of the pole's damp-ening coefficient and the friction coefficient.

**5** and **6** introduce the exponential decay of the pole to model air resistance and friction of the pole bearings. Further reducing the mismatch between simulation and RW Cartpole leads to another improvement in the performance ratio. **5** uses the sigmoid approximation for the cart friction, **6** the step function. Both show noticeable improvements over **2** and **4**, respectively.

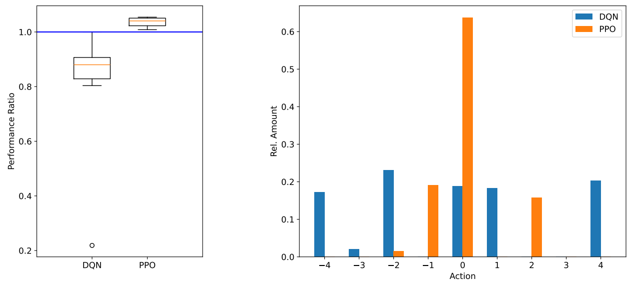
As described in [28], [29], DR can lead to a more robust sim-to-real transfer. We can DR at multiple locations inside our framework. First, we can apply DR to the actual state of the Cartpole, i.e., $(x, \dot{x}, \phi.\dot{\phi})$. Furthermore, we can also randomize the static parameters of the simulation. In the case of our introduced models for friction, we can apply randomization to the cart's coefficient of friction and the pole's dampening coefficient. When considering DR, it is always necessary to keep constraints such as boundary values in mind that apply to the considered quantity. For example, our static friction models need the friction force $\mu * F_{norm} > 0$ because a negative friction force would lead to a positive feedback loop of the cart. Similarly, we want the damping coefficient of the pole to be smaller than one. Otherwise the oscillations would not be dampened but rather amplified. Due to the constraints, we cannot apply infinitely wide distributions to sample from. Instead, we need semi-infinite or finite distributions. We tested uniform distributions with min and max within $\pm x$ percent from the measured values. Fig. 4 shows our result using DR.

Perturbating the pole's dampening coefficient mainly affects the performance ratio values (shift of the boxes on the y-axis). However, it barely affects the variability in the performance ratio (size of the boxes). For a perturbation of 6%, the box lies nearly symmetric around the ideal ratio of one. The randomizations at the start of each simulation run and the inherent random effects of the RW Cartpole setting cause the variability leading to the observable height of the box. The symmetry around the ideal ratio suggests an even closer match between the agent's performance in simulation and RW settings. Increasing the randomization of the pole's dampening coefficient even further, e.g., 12%, can significantly decrease the sim-to-real performance. Not only present on the RW Cart-pole but also within the simulation, the agent shows difficulties adapting to different simulation settings if parameters vary in unsuitable ranges. A perturbation of 3% in the coefficient of friction of the cart shows a slight increase in performance similar to the previous experiment of perturbation of the pole's dampening coefficient. However, further perturbations reduced the performance significantly.

DR can lead to a better generalization of the agent to the environment. We find that it is a valuable tool to close the reality gap. DR is especially beneficial if exact modeling of the underlying physics is unfeasible, too expensive to simulate, or too cumbersome to measure.

Finally, we compare the two realms of RL, off- and on-policy algorithms. As a representative of off-policy algorithms, we use DQN. For on-policy algorithms, we chose PPO. Regarding PPO, we found it very challenging to identify suitable hyperparameters to obtain a converging learning curve for the swing-up task. In the following, we only consider the balancing task for a fair comparison of our methods and existing work. While the performance difference of on- and off-policy agents can be more significant for the swing-up task, observing the balancing task can give more insight into possibly more fine-tuned control strategies.

In Fig. 5a, we observe a significant performance difference between the DQN and PPO algorithms. Fig. 5b shows the relative amount of the agents' actions and further explains the visible difference. While PPO decides to apply no force most of the time with few minor corrections, DQN tends to oscil-late between actions. Oscillating between neighboring actions could be helpful to average the applied force between available action values. However, our trained DQN ignores actions -3,-1, 2, and 3 almost entirely, resulting in a performance decrease.




(a) Performance ratio for DQN and PPO.

(b) Control strategies of DQN and PPO.

Fig. 5. Comparison of DQN and PPO for the Cartpole balanc-ing task.

We assume that the performance difference between off- and on-policy algorithms is due to three main factors: robustness to distribution shifts, stability of training, and adaption to uncertainty in the environment. On-policy algorithms could be advantageous in applications like the Cartpole problem since they learn from the most recent data. Thus, they may suffer less from distribution shifts, divergence or instability during training, or uncertainties in the environment.

## VI. Conclusion

The growing relevance of ML in research and our everyday lives increases the need to test and evaluate algorithms not just in idealized simulation environments but also in RW applications, particularly in safety-critical settings such as autonomous driving or UAV control. The Gymnasium Cartpole environment is one of the de-facto standards of RL bench-marking for simulations. Given our RW Cartpole experimental setup, we propose an enhanced Cartpole environment as a new benchmark for sim-to-real transfer or physics-informed RL. It can be used as a drop-in replacement for the original environment. We have demonstrated that we closely match the RW dynamics of our Cartpole setting with our modifications and thus successfully address the reality gap.

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